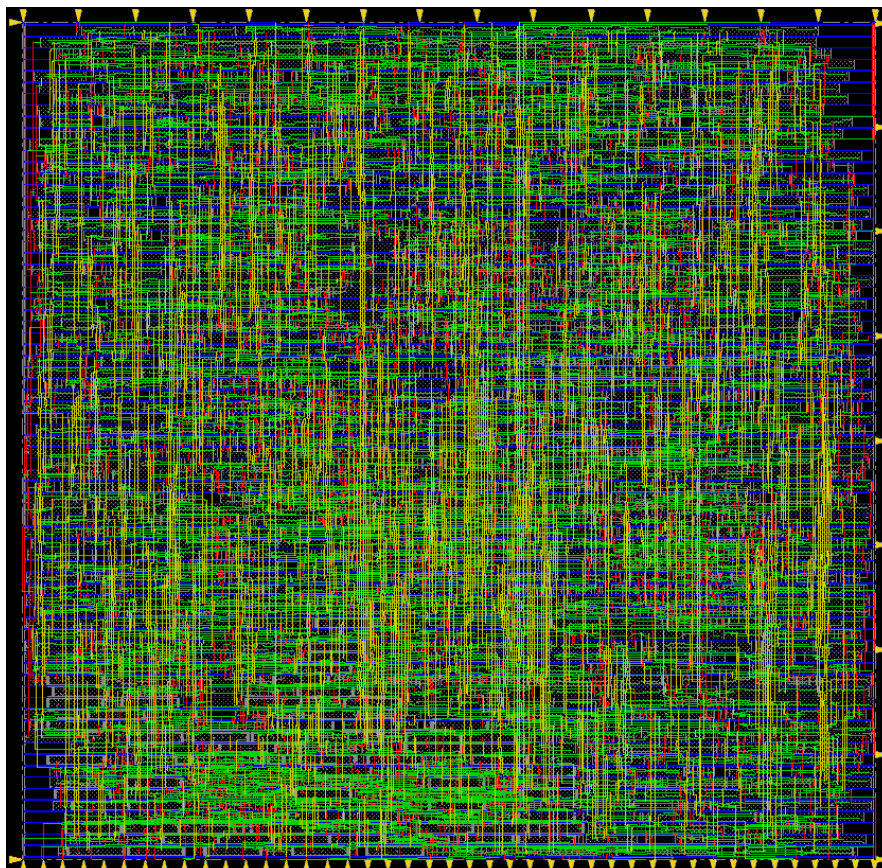


2A Internship Report

Design and Evaluation of a Radiation-Tolerant Set-Associative Cache with an ECC Mechanism in an ASIC Design Flow

2025-2026

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A Presentation of the Internship Location

A.1 The Host Laboratory

The internship took place within GMicro (Grupo de Microeletrônica), more specifically in the laboratory of the Federal University of Santa Maria (UFSM, Brazil). The Santa Maria Design House (SMDH) is also located there and is the structure dedicated to the design of integrated circuits for aerospace, defense, and communications applications. The laboratory is part of the Brazilian federal CI-Brasil program, which aims to develop a national semiconductor industry. It is headed by Professor João Baptista dos Santos Martins, who also supervised this internship.

A.2 Autonomy and Integration within the Team

The internship topic was part of a research area on which no other member of the laboratory was actively working during this period (fault tolerance and radiation-hardened circuits). The internship therefore took place with a high degree of autonomy: the definition of the technical objectives, specifications, and work schedule was left to my initiative, subject to occasional validation by my supervisor.

A colleague from the laboratory, Elias De Almeida Ramos, nevertheless took some time at the beginning of the internship to present his previous work. In particular, he is a co-author of IEEE publications on linear algebra-based ECC codes as well as on the RHBD library developed in the laboratory. These two documents served as the main references for the state of the art presented in this report (§ C)[1][2][3]. His current work now focuses on other topics (SAT problem solving, clustering) that are not directly related to this internship.

A.3 Work Environment and Conditions

An individual workstation was provided, along with access to a server hosting the laboratory's tools and licenses (Cadence Genus, Innovus). The latter is accessible via SSH and can be configured as a remote host from a local development environment (VS Code). VPN access also made it possible to work remotely. The configuration files for the RHBD library were provided by another member of the laboratory, and their integration into the design flow was then carried out independently (§ D.4).

Working hours were not subject to any formal constraints, with each team member managing their own schedule. I therefore chose a regular schedule of approximately 7 hours per weekday, occasionally supplemented by work on weekends.



Figure 1: Workstation within the GMicro Laboratory

B Introduction

B.1 Project Context and Challenges in the Space Industry

Satellites and spacecraft carry a large number of microelectronic devices. They are designed to operate with very high reliability in particularly harsh environments where they are exposed to high levels of radiation. Most of these devices consist of integrated circuits (ICs), such as FPGAs (Field Programmable Gate Arrays), ASICs (Application-Specific Integrated Circuits), or dedicated microprocessors.

Another challenge related to the space sector is the reduction of payloads, which leads to a decrease in the size and weight of equipment. Today, most analog and digital systems can be integrated into a single chip or package (System-on-Chip: SoC). Although the design processes are not yet fully standardized, this approach is expected to become widespread in the field of space microelectronics in the short and medium term.

B.2 Problem Statement: Vulnerability of Cache Memories in Radiation Environments

In the space environment, particle emissions are more frequent, which can disrupt circuit operation. These interactions between radiation and silicon can, in particular, cause changes to the data stored in memory over time (bit flips).

As will be discussed later, these phenomena are amplified by the reduction in technology nodes. This is why advanced technologies require more sophisticated protection techniques such as ECC (Error Correcting Code) or RHBD (Radiation Hardening by Design) techniques.

Due to technological constraints, low production volumes, and limited demand for integrated circuits intended for space applications, RHBD techniques using a conventional CMOS process have become particularly attractive in recent years. This approach makes it possible to mitigate the effects of radiation without modifying the manufacturing process.

With this in mind, an RH (Radiation Hardened) library using 180 nm silicon technology was developed at the Gmicro laboratory. It relies in particular on the use of Guard Rings and the application of the Enclosed Layout Transistor (ELT) technique. This library includes simple digital logic gates as well as more complex cells. The radiation-hardened digital cells that have been developed have demonstrated tolerance to energetic particles of solar origin with energies of up to 100 MeV.

B.3 Internship Objectives and Expected Deliverables

The internship is divided into two phases. First, the objective is to study the types of radiation that an integrated circuit can be exposed to, particularly in space. Then, the objective is to study and implement techniques for protecting against radiation-induced errors in a memory subsystem.

The work focuses on the SystemVerilog design of a set-associative cache architecture integrating an error correction mechanism (ECC), as well as its evaluation within a complete ASIC flow. Particular attention will be paid to the comparison between an implementation using a standard library and an implementation using an RHBD library. The goal will then be to evaluate the trade-offs between performance, occupied area, power consumption, and fault tolerance.

C State of the Art and Theoretical Background

C.1 Effects of Radiation on Semiconductors

C.1.1 Cumulative Effects (Cumulative Effects)

TID (*Total Ionizing Dose*) effects are caused by ionizing radiation (X-rays, gamma rays, electrons, protons). As they pass through a transistor, they create electron-hole pairs in the insulating regions. The holes remain trapped at the interface between the gate oxide and the silicon[4], resulting in three main consequences : a shift in the threshold voltage, an increase in leakage currents, and a reduction in transconductance (slower transistors)[4].

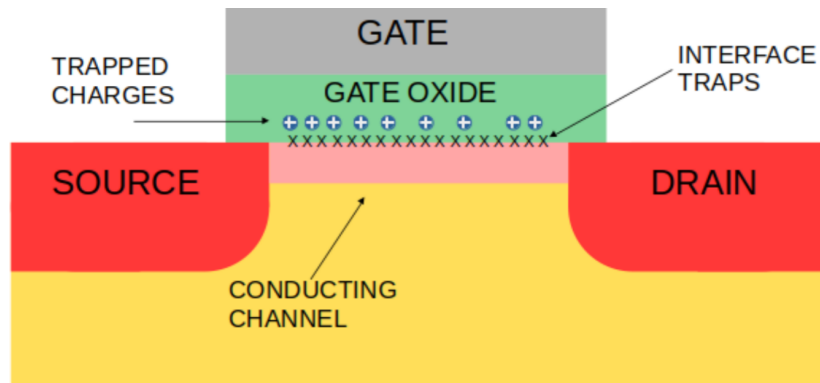


Figure 2: Trapping of positive charges in the gate oxide[4]

DD (*Displacement Damage*) effects are caused by heavy non-ionizing particles (neutrons, protons, heavy ions). They displace atoms in the silicon crystal lattice and create structural defects[4]. This mechanism has little effect on MOS transistors and will therefore not be discussed in further detail in this report.

C.1.2 Non-Destructive Single-Event Effects (SET and SEU)

SEE (*Single Event Effects*) result from the impact of a single ionizing particle (heavy ion, proton, neutron...). They locally generate electron-hole pairs that disrupt circuit operation[4]. Two non-destructive effects are particularly relevant to a memory subsystem :

- **Single-Event Transient (SET)** : a purely analog transient voltage or current spike that propagates through the combinational logic, making its detection difficult[4].
- **Single-Event Upset (SEU)** : a change in the logical state of a memory cell (*bit flip*). When a sufficiently energetic particle corrupts several adjacent bits, this is referred to as a *Multiple-Bit Upset* (MBU)[4].

It is this latter phenomenon, which is directly related to the reliability of a cache memory, that motivates the implementation of an error correction mechanism (§ C.2.2). Furthermore, as modern technology nodes become smaller, the charge required to change the state of a transistor is reduced[4]. In addition, memory cell density increases, which promotes MBU/MCU (Multiple-Bit Upset/Multiple Cell Upset)[1].

C.1.3 Destructive Single-Event Effects (SEL)

SEE can also cause permanent failures. **Single-Event Latch-Up (SEL)** is the most critical effect in CMOS technology : the deposited charge triggers a parasitic thyristor structure between the power supply (V_{DD}) and ground (GND), which becomes self-sustaining and creates a direct short circuit, destroying the component through overheating if the power supply is not shut down quickly enough[4].

Two other destructive effects are described in the literature (*Single-Event Burnout* and *Single-Event Gate Rupture*), but they concern power devices operating under high blocking voltages. They are therefore not representative of the digital MOS transistors used in this project and will not be discussed here. SEL, however, remains a direct threat to the designed cache, which justifies the use of the hardening techniques presented below.

C.2 Hardening and Mitigation Techniques

C.2.1 Radiation Hardening by Design (RHBD)

Radiation-Hardened By Design (RHBD) consists of modifying the geometry of components during the design phase without changing the manufacturing process. It makes it possible to mitigate the effects of SEL and TID. Two layout techniques are used in the laboratory's library.

Guard Rings.

This technique reduces the sensitivity of CMOS circuits to latch-up[6]. It consists of placing a heavily doped region (P+ or N+) around the transistors, with this region connected to a power supply. It acts as a collector by quickly capturing the carriers injected into the silicon through a low-resistance path to the power supply contacts. The voltage drop in the N and P wells is therefore reduced[6]. The parasitic thyristor responsible for latch-up is thus less likely to trigger because its junction is more weakly biased[6].

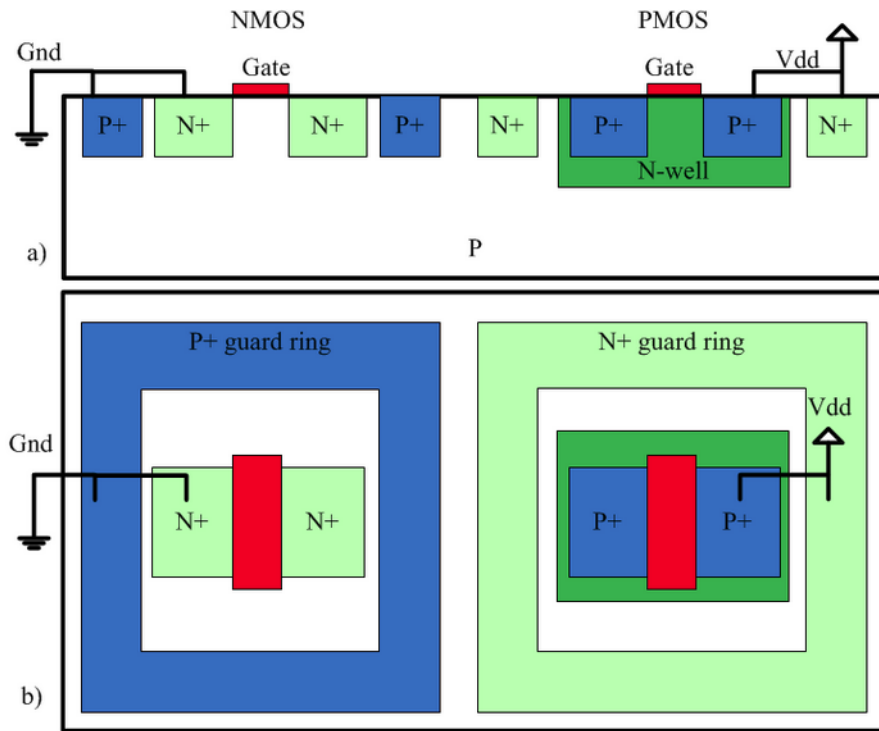


Figure 3: Cross-section of an NMOS and a PMOS with their respective guard rings (adapted from [8])

Enclosed Layout Transistors (ELT).

Trapped charges caused by irradiation increase leakage currents and degrade transistor performance [5]. This issue is reduced by decreasing the gate oxide thickness, but persists at the level of the isolation trench (STI: Shallow Trench Isolation) separating the transistors[5]. The ELT (*Enclosed Layout Transistor*) technique addresses this issue through a ring-shaped gate geometry, with the source (or drain) placed inside and the opposite electrode outside. The ends of the

channel are therefore no longer in contact with the STI, which eliminates any parasitic leakage path that could appear between the drain and source after irradiation[5].

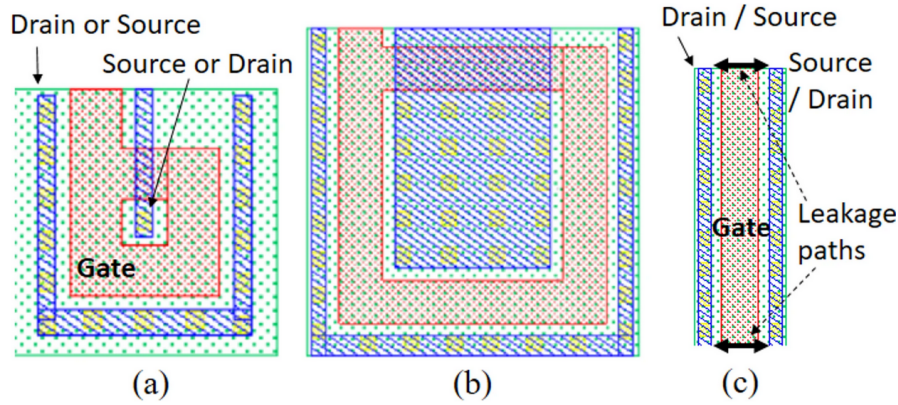


Figure 4: Layout of different ELTs (a, b) and a conventional transistor with leakage paths induced by TID (adapted from [7])

C.2.2 Architectural-Level Mitigation: Error-Correcting Codes (ECC)

Another approach to radiation protection is implemented at the architectural level : it consists of adding error correction logic (ECC: *Error Correcting Code*) to detect and correct *bit flips* in memory[2]. The most widely used example is the Hamming SECDED code, which detects 2-bit errors and corrects 1-bit errors.

An ECC Based on Linear Algebra.

The UFSM laboratory (Santa Maria Design House) has developed an alternative approach to conventional codes (Hamming, BCH). It is based on linear algebra over the finite field $GF(2)$ [1][2]. The data word d is viewed as a vector encoded by a bijective linear transformation A into a redundancy word r of the same size :

$$r = A(d) \quad (1)$$

The word stored in memory is the concatenation of the two vectors, $C = (d, r)$, resulting in a redundancy rate of 100 % for a $2n$ -bit encoded word.

The purpose of the transformation A is to diffuse the information contained in d : each bit of r depends on several bits of d (mixing property), thereby increasing the distance between valid codewords and improving error detection[1][2]. Unlike a Hamming code, the redundancy bits are therefore not independent parity bits, but a complete representation of the original word (see the numerical example in Appendix H.2).

Decoding by Syndrome Comparison.

Upon reading, the received word (d', r') may have undergone bit flips. The decoder recomputes $A(d')$ and $A^{-1}(r')$ and compares these results with the read values to form two error vectors, called syndromes :

$$\text{syndrome_data} = d' \oplus d_{reconstructed} \quad \text{syndrome_red} = r' \oplus r_{calculated} \quad (2)$$

Thanks to the diffusion property of A , a cluster of localized errors (MCU) strongly affects one of the two syndromes while leaving the other with a minimal weight, allowing for a simple decision rule based on the number of bits set to 1 in each syndrome :

- if $\text{weight}(\text{syndrome_data}) < \text{weight}(\text{syndrome_red})$, the redundancy block is considered more reliable : the reconstructed data $d_{reconstructed}$ is selected, thereby correcting d' ;

- otherwise (or in the event of a tie), the raw data d' is retained and transmitted as output.

This ability to handle multiple errors (MCU) distinguishes this ECC from traditional codes. This is even more relevant as technology nodes shrink, increasing the probability that a single particle will affect several neighboring cells[1].

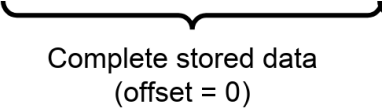
D Specification and Design of the Architecture in SystemVerilog

D.1 Set-Associative Cache Architecture

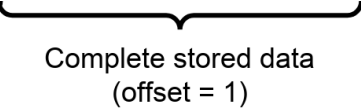
D.1.1 Architectural Choices and Parameterization

The cache designed in this project is a 2-way set-associative cache whose overall architecture is based on the principles covered in the *Processor Architecture II* course. It consists of 8 sets, each containing two ways. Each line stores a 32-bit word of useful data, divided into two groups of two bytes that can be individually accessed through an offset bit; the first byte is used to store the data and the second the associated redundancy.

	[36 : 33]	[32]	[31 : 24]	[23 : 16]	[15 : 8]	[7 : 0]
Sets	tag	Valid	Data 1	Redundancy 1	Data 2	Redundancy 2
0	0011	0	32	188	4	55
	1010	1	7	253	95	64
1	0000	1	45	121	65	76
	1110	1	127	252	34	43
...	...	...	...	...	...	...
7	0110	0	12	152	57	77
	1010	1	255	180	62	176



Complete stored data
(offset = 0)



Complete stored data
(offset = 1)

Figure 5: Cache memory organization

The CPU address is encoded on 8 bits and is divided according to the following structure :

- **Tag** (4 bits) : identifier of the line in main memory,
- **Index** (3 bits) : selection of the set in the cache,
- **Offset** (1 bit) : selection of the byte within the line (lower or upper byte).

This partitioning into structures significantly improves the readability and maintainability of the code. It allows direct access to the fields of an address or a cache line rather than manipulating raw bit vectors (structure details in Appendix H.1).

Finally, the cache must be able to handle the 4 cases it may encounter: the CPU may request a data read (READ) or a write (WRITE), and the requested data may already be present in the cache (hit) or not (miss).

The controller operates only on useful data (two bytes per line), while the SRAM physically stores the two additional bytes of ECC redundancy along with the data. Encoding is performed during a write operation by an ECC_encoder block, while decoding (with error correction) is performed by two ECC_decoder blocks (one per way).

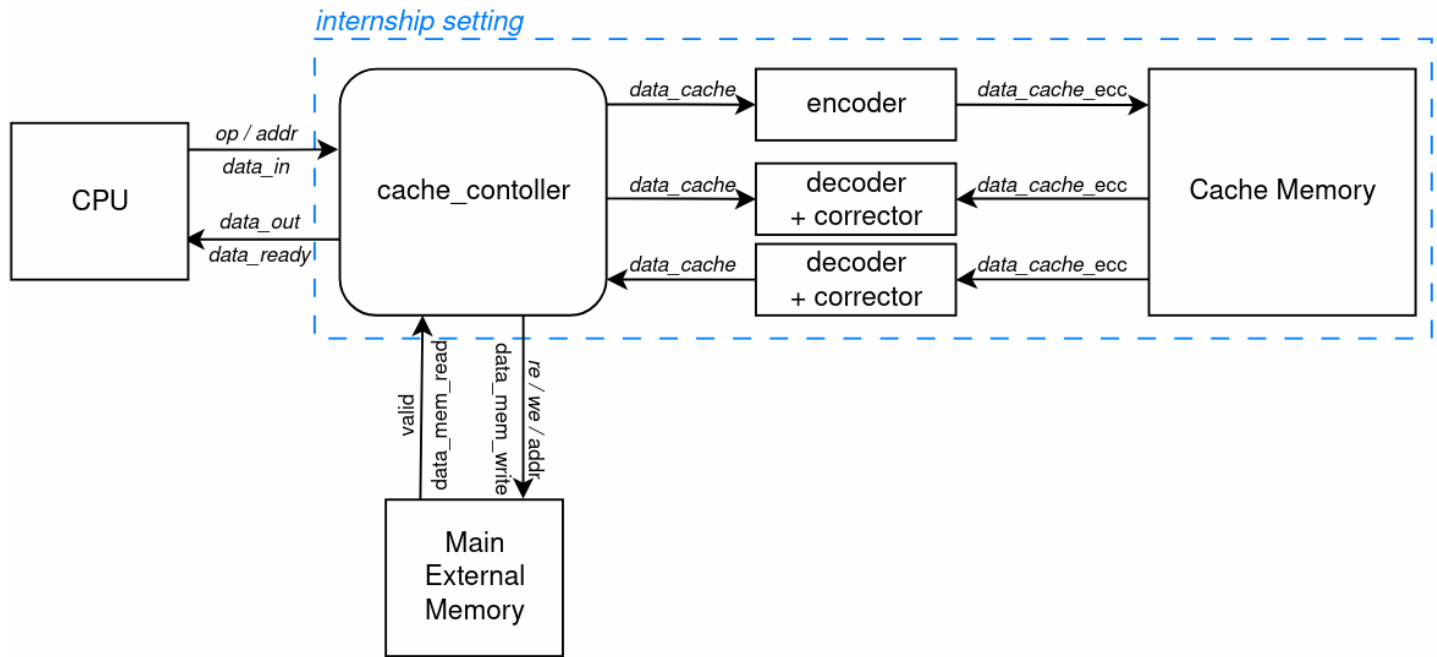


Figure 6: Block diagram of the complete system: cache controller, SRAM, and ECC blocks

D.1.2 Finite State Machine (FSM) and Control Logic

The "brain" of the cache controller is a finite state machine (FSM, *Finite State Machine*). It is a sequential circuit that controls the behavior of the system based on its current state and input signals by carrying out the necessary actions step by step. The designed FSM consists of five states :

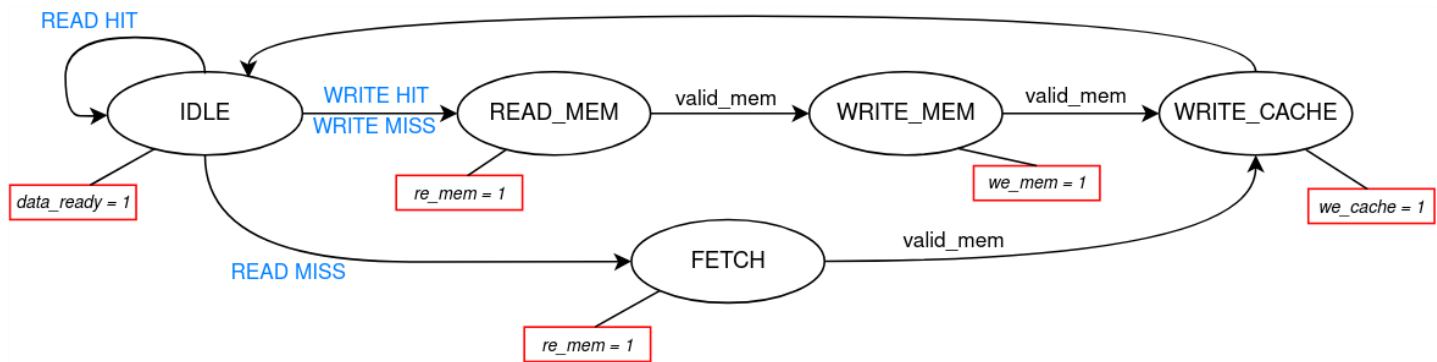


Figure 7: State diagram (FSM) of the cache controller

- **IDLE** : idle state. The controller continuously evaluates the hit0 and hit1 signals, calculated by comparing the tag of the CPU address with the tags of the two ways read from the SRAM. In the event of a *read hit*, the data is immediately provided to the CPU. Any write or *read miss* triggers a transition to an active state.
- **FETCH** : triggered on a *read miss*. The controller issues a read request to main memory and waits for the data, which is then transmitted to the CPU before transitioning to WRITE_CACHE.
- **READ_MEM** : triggered by a write (hit or miss). To respect the byte-level granularity of the CPU when dealing with a 16-bit line, the existing line is first read from main memory so that only the targeted byte can be modified while preserving the other.

- **WRITE_MEM** : once the line has been reconstructed with the new CPU data, the controller writes it to main memory and waits for the acknowledgment.
- **WRITE_CACHE** : writing the updated line to the cache SRAM. The implemented replacement policy is a true LRU (*Least Recently Used*) policy, encoded on one bit per index, indicating the least recently used way.

The FSM implements a *write-through* policy with *write-allocate* : every write is systematically propagated to main memory and the corresponding line is always loaded into the cache. This symmetry simplifies the FSM since the execution path is identical for a *write hit* and a *write miss*.

D.1.3 Issue Encountered: SRAM Pipeline and Synthesizability

An issue arose during the integration of the SRAM: the controller calculates `hit0` and `hit1` combinationally from the SRAM outputs. However, a synchronous SRAM (required for ASIC synthesis) has a one-cycle latency between the presentation of the address and the availability of the data at the output. The controller was therefore evaluating hits on data corresponding to the address from the previous cycle, resulting in false misses. The adopted solution introduces an `addr_cpu_reg` register that delays the CPU address by one cycle: the hit signals are calculated using this register, which is perfectly aligned with the SRAM response, while the address presented to the SRAM remains combinational to anticipate the read for the following cycle.

D.2 Integration of the Error Correction Mechanism (ECC)

D.2.1 Hardware Implementation of the ECC

The hardware architecture of the ECC directly implements the code presented in § C.2.2, developed by the laboratory. Encoding and decoding are each implemented as a single high-level combinational block :

- `ECC_encoder` : takes the two data bytes of a line as input, calculates the associated redundancies using the XOR equations of the code, and assembles the complete structure stored in SRAM (data, redundancies, tag, valid).
- `ECC_decoder` : performs the inverse operation with error correction. One instance is associated with each cache way.

Correction Logic in `ECC_decoder`.

Correction is performed purely combinationally. For each of the two bytes of the read line, the module recalculates a reconstructed data value and a reconstructed redundancy value, then compares these recalculated values with the read values using two 8-bit syndromes :

- `syndrome_data = data_in  $\oplus$  data_reconstructed` : difference between the raw data read and the reconstructed data,
- `syndrome_redun = redun_in  $\oplus$  redun_reconstructed` : difference between the raw redundancy read and the redundancy recalculated from the read data.

The correction decision is based on a strict comparison of the Hamming weights of these syndromes, evaluated in hardware using the `$countones` operator :

$$\text{data_cache.data}_i = (\text{weight_data}_i < \text{weight_redundancy}_i) ? \text{data_reconstructed}_i : \text{data_in}_i \quad (3)$$

If the number of erroneous bits identified in the data is strictly lower than that identified in the redundancy, the multiplexer outputs the corrected data. Otherwise (or in the event of a tie), the raw data read is retained.

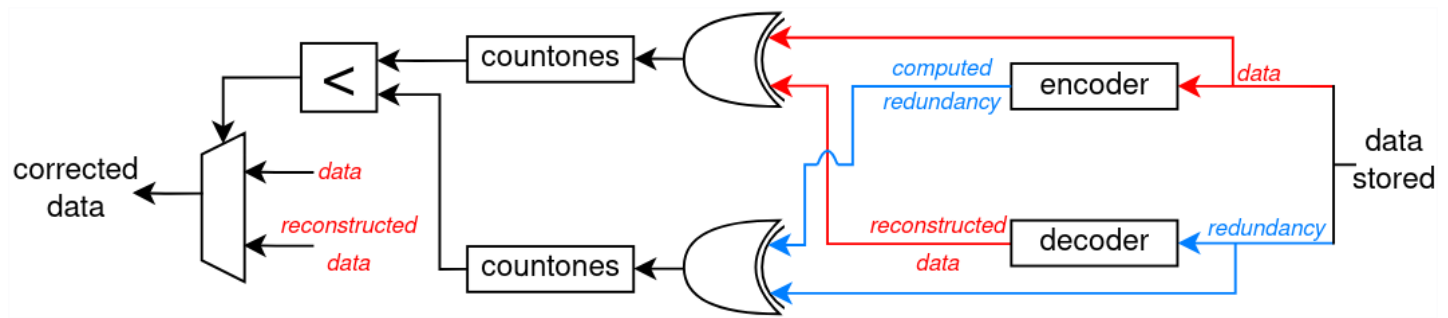


Figure 8: Block diagram of the decoder architecture with correction (only one way shown)

D.3 Functional Validation Strategy and Testbenches

The design was validated by testing each new part individually.

Cache Controller Validation (cache_controller_tb).

The controller was validated independently using behavioral models for the SRAM (with inputs driven directly) and for the main memory (with a configurable latency of several cycles). Six scenarios were verified : READ hit with offset 0 or 1, complete READ miss with loading from main memory, hit after miss (consistency after writing to the cache), WRITE hit, and WRITE miss. The proper operation of the LRU replacement policy was also verified during these scenarios.

ECC Validation (ECC_tb).

The encoder is verified by comparison with analytically calculated redundancy values. The decoder is tested through controlled error injection : bits are deliberately corrupted in the data or redundancy field, and the decoder must restore the original value.

System Validation (cache_ECC_tb).

This testbench integrates the controller, SRAM, encoder, and two decoders. A `write_memory` task abstracts the pre-loading of the cache memory by handling the complete write sequence through the controller (READ_MEM → WRITE_MEM → WRITE_CACHE). The same six scenarios were replayed in this integrated context.

D.4 Development Environment and Implementation of the ASIC Flow

The laboratory provided a complete digital design flow (from RTL to GDSII) managed by a Makefile and an already functional file hierarchy. However, this flow was configured exclusively for the academic standard library (45 nm PDK). Part of the work therefore consisted in extending this environment to also support the **RHBD** library (180 nm, XFAB XH018) developed internally, and then resolving the placement and routing issues specific to this library.

D.4.1 Multi-PDK Support

The provided flow was modified to dynamically load a technology configuration file according to an input variable (TECH=rhbd or TECH=std). It encapsulates all the files associated with a technology: .lib (delays, *Slow/Fast/Typical* corners), .lef (cell and metal layer geometries), and capacitance tables (capTbl). The same Makefile can therefore be used to easily perform the ASIC flow steps on either technology.

D.4.2 Placement and Routing Challenges Specific to the RHBD Library

The integration of the RHBD library under Innovus raised two physical issues related to the large size of the hardened cells and the constraints of the 180 nm node.

Fixing the Hardened D Flip-Flops.

The hardened D flip-flop (RH DFF) has a significantly larger physical footprint than the rest of the logic due to its design. During the global optimization phases, Innovus attempted to move these cells and caused placement rule violations for these cells. A *dont_touch* constraint was therefore applied in the TCL scripts to fix their placement.

Simplification of the Power Distribution Network (PDN).

Since the technology has only 4 metal layers, reserving a complete vertical layer for the power rails quickly saturated the available space. As the memory block is of moderate size (32 useful bytes), the vertical rails were removed in favor of horizontal-only distribution. The voltage drop (*IR Drop*) remains moderate in this case (see Appendix H.3).

E Evaluation of the Memory Subsystem: Robustness and PPA-F Trade-offs

E.1 Validation Strategy

The ideal evaluation of radiation robustness relies on physical tests (heavy-ion, proton beams). The laboratory does not have such equipment, and no circuit was fabricated during this internship. The evaluation therefore relied on two simulation approaches :

- a **Python Monte Carlo campaign**, to statistically quantify the intrinsic performance of the ECC against *bit flips* (protocol detailed in Appendix H.4).
- a **verification using a SystemVerilog testbench**, to confirm the faithful reproduction of the algorithmic model in RTL code .(protocol detailed in Appendix H.5).

E.2 Fault Robustness: ECC Performance

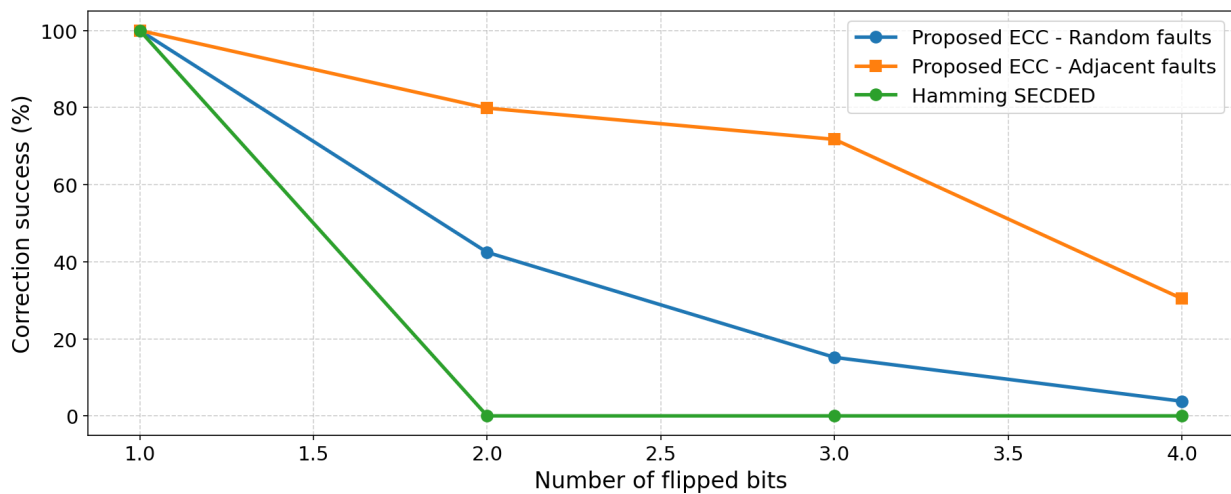


Figure 9: Correction success rate of two ECCs as a function of the number of faults

The Python simulations highlight a strong asymmetry depending on the fault distribution.

For the proposed ECC, with two purely random faults (two independent SEUs), the correction rate drops to **42 %**. In contrast, the rate rises to **80 %** for two adjacent faults (MCU). This shows that the diffusion (*mixing*) property of the linear transformation over $GF(2)$ effectively disperses a localized error pattern and therefore enables effective correction against MCUs. This property is highly relevant for the targeted application, as energetic particles predominantly generate faults grouped on contiguous cells.

Beyond two faults, the 8-bit format reaches its structural limits : the correction rate drops to **31 %** for 4 adjacent errors and to **5 %** in random mode. The errors saturate the redundancy equations and cause the syndromes to lose their asymmetry, which misleads the decision logic based on syndrome weight.

This limitation confirms that the algebraic code must imperatively be combined with physical layout protections (Guard Rings, ELT, § C.2.1) to prevent the occurrence of fault clusters larger than 2 bits, the latter limiting the effective size of charge deposition.

It is nevertheless worth noting that the performance of the proposed ECC is significantly higher than that of a traditional SECDED (Single Error Correction, Double Error Detection) ECC for only 3 additional redundancy bits (5 $\rightarrow$ 8).

Consistency Between the Software Model and Hardware Implementation.

A **cross-validation** strategy was implemented to verify that the correction rate obtained by the Python model is effectively reproduced by the hardware implementation. The same fault vectors (data, redundancies, expected result after correction) generated by the Python model are exported to a text file, then replayed directly on the ECC_decoder module by a dedicated SystemVerilog testbench. The testbench then compares the hardware output with the expected value (methodology detailed in Appendix H.5).

For 10 000 vectors per configuration, the correction rates obtained in hardware faithfully reproduce those of the software model : 79,70 % versus 80 % for two adjacent faults, and 30,61 % versus 31 % for four adjacent faults. The small differences observed are solely explained by the random selection of test vectors being different between the two campaigns. This consistency confirms that the RTL module faithfully implements the syndrome-weight-based decision rule of the algorithmic model.

E.3 Area

Table 1 breaks down the area occupied by cell category at the maximum frequency of each architecture.

Table 1: Comparative area analysis and equivalent logic complexity (GE) at $F_{\max}$

Category of cell	Std Cell 45 nm (335 MHz)			RHBD 180 nm (80 MHz)			Factor increase
	Area	%	GE	Area	%	GE	
Flops (Sequential)	5 739,44 μm^2	64,6 %	5 594	339 257,35 μm^2	64,3 %	5 770	$\times$ 59,1
Combinational	2 849,20 μm^2	32,1 %	2 777	150 274,37 μm^2	28,5 %	2 556	$\times$ 52,7
Inverters	180,92 μm^2	2,0 %	176	25 675,08 μm^2	4,9 %	437	$\times$ 141,9
Buffers	114,23 μm^2	1,3 %	111	12 131,97 μm^2	2,3 %	206	$\times$ 106,2
Total Core	8 883,79 μm^2	100 %	8 658	527 338,77 μm^2	100 %	8 968	$\times$ 59,4

At first glance, the enormous difference in area between the two technologies is immediately apparent. However, the $\times 59$ factor observed in total area can be fully explained : in terms of *gate count* (GE), the raw logic complexity increases by only 3,5 % (8 658 $\rightarrow$ 8 968 GE), with the difference mainly corresponding to a few buffers inserted by Innovus to balance routing distances in

180 nm.

The increase in physical area is therefore attributable to the intrinsic geometrical cost of hardening (Guard Rings, ELT, § C.2.1) and the change in technology node (180 nm versus 45 nm), rather than to a difference in logic complexity.

The relative distribution by category also remains stable from one technology to the other (~64 % sequential, ~30 % combinational), with inverters and buffers being proportionally more numerous in RHBD due to longer paths that need to be balanced.

The ECC (encoder + decoders) alone represents $1\,373,81\,\mu\text{m}^2$ in the standard library, corresponding to **15,46 %** of the total active area of the cache.

E.4 Timing Performance

Table 2: Maximum frequencies obtained according to the cell library

Library	Min. Period	F_{max}	Slack (WNS)
Standard (45 nm)	2,985 ns	335 MHz	0,016 ns
RHBD (hardened)	12,500 ns	80 MHz	0,166 ns

In both libraries, the critical path has the **same structure** : memory output flip-flop → combinational logic of the ECC decoder → final multiplexing in cache_controller.

It never passes through the FSM: the bottleneck is therefore exclusively combinational and localized in the ECC. The number of stages traversed is comparable between the two libraries (24 cells in the standard library, 19 in RHBD), showing that the frequency gap does not result from a different logic depth but from the **unit delay per cell** : the delay of the memory output flip-flop increases from 0,271 ns to 1,009 ns ($\times 3,7$) between standard and RHBD, while the gates used for syndrome computation increase from 50–80 ps to 400–700 ps, due to the higher internal gate capacitances of the hardened cells.

This penalty, repeated at each stage of the ECC decoder directly connected to the SRAM output, explains most of the difference in maximum frequency (335 MHz versus 80 MHz).

E.5 Energy Consumption

Table 3: Post-route power breakdown according to the cell library

Tech.	Internal	Switching	Static	Total	% Static
Standard 45 nm (0,9 V)	0,708 mW	0,184 mW	0,186 μ W	0,892 mW	0,021 %
RHBD 180 nm (1,62 V)	41,43 mW	5,187 mW	2,088 μ W	46,62 mW	0,004 %

The ~52-fold factor in total power consumption is explained by two cumulative effects. The supply voltage is nearly doubled (1,62 V versus 0,9 V, with dynamic power varying as V_{DD}^2). In addition, RHBD cells are intrinsically more power-hungry (larger internal gates), which explains the ~58-fold factor observed in internal power. As with timing, the hardened sequential elements dominate the power breakdown (70,85 % of the total in RHBD versus 65,84 % in standard), with the ECC logic accounting for only 27,54 % to 34,16 % of the power consumption.

The relative share of *leakage* is, however, lower in RHBD 180 nm (0,004 %) than in standard 45 nm (0,021 %). This is because the reduction in technology node increases leakage currents through tunneling across the gate oxide, an effect that the 180 nm node is less susceptible to than the 45 nm node. Furthermore, the elimination of inter-transistor leakage paths through Guard Rings (TID) contributes to better behavior of the RHBD library over the lifetime of a space mission.

E.6 Trade-off Between Placement Density, Frequency, and Area

Beyond the choice of library, the target placement density itself constitutes an engineering parameter that must be balanced. The densities ultimately selected for placement and routing are 90 % for the standard library and 89 % for the RHBD library. This choice results from a trade-off : increasing the density brings the cells closer together and reduces area, power, and *setup slack*, but conversely degrades the *hold slack*. It is also observed that beyond a certain threshold, congestion becomes so high that Innovus can no longer route the circuit, resulting in violations during DRC checks. The selected value therefore corresponds to the highest density for which routing remains possible without excessive hold degradation.

To illustrate this approach, two syntheses of the standard library were compared at 90 % and 25 % density at a period close to F_{max} .

Table 4: Evolution of PPA metrics between 90 % and 25 % density (45 nm std cell)

Metric	Evolution (90 % → 25 %)	Trend
Setup slack (WNS)	-0,006 ns	Degradation
Hold slack	+0,003 ns	Improvement
Area	+19 μm^2 (+0,2 %)	Slight degradation
Total power	+9 μW (+1,1 %)	Slight degradation

The results confirm the expected trends : a higher density brings the cells closer together, shortens the routing paths, and improves *setup slack* as well as area and power (shorter paths, lower conduction losses), but degrades the *hold slack*. This is because the propagation times between adjacent flip-flops become shorter than the required minimum delay. For this moderately sized design, the effect remains marginal on area and power (< 1,5 %), but nevertheless illustrates the reasoning followed to determine the final density : push it as high as possible as long as hold remains controlled and routing completes without violations.

E.7 PPA-F Trade-off Summary

Table 5: Comparative summary of performance and robustness (PPA-F Trade-off)

Metric	Standard (45 nm)	Hardened (180 nm RHBD)
Maximum Frequency (F_{max})	335 MHz	80 MHz
Total area	8 883,79 μm^2	527 338,77 μm^2
Total power consumption	0,892 mW	46,62 mW
TID tolerance (dose)	Low	Better immunity
SEL tolerance	Critical (destructive)	Better immunity
		High
SEU / MCU tolerance	Low (raw SRAM)	(> 70 % up to 3 bit flips in MCU)

Note : only SEU/MCU tolerance was quantitatively characterized (§ E). TID and SEL tolerance are indicated qualitatively, based on the known properties of the RHBD library documented by the laboratory, without verification through testing or simulation as part of this work.

Conclusion on Architectural Choices.

Radiation hardening imposes a degradation of the PPA trade-off in favor of fault tolerance : the increase in area ($\times 59$) directly results from the geometry of the RHBD cells (Guard Rings, ELT), while the increased power consumption ($\times 52$) is explained by the same area increase combined with the higher supply voltage. The resulting frequency reduction (335 MHz → 80 MHz) remains secondary compared with the reliability requirements of the space environment. The proposed architecture thus illustrates the benefit of a two-level protection approach: the ECC corrects

transient logical alterations (SEU, MCU up to 2 adjacent bits, with an 80 % measured success rate), while the hardened layout protects the silicon against cumulative and destructive effects (TID, SEL) and physically limits the extension of fault clusters that the ECC alone can no longer correct beyond 2 bits.

F Conclusion

F.1 Technical Conclusion

The objective of this internship was to design and evaluate a set-associative cache integrating an error correction mechanism (ECC). Once this stage was completed, the focus shifted to comparing its implementation using a standard library with an implementation using a radiation-hardened (RHBD) library.

The proposed architecture is based on a 2-way cache controller driven by an FSM and associated with an original ECC developed at the GMicro laboratory.

The simulation campaigns (Python model and SystemVerilog testbench) confirmed the significant benefits of this ECC : with a moderate redundancy overhead (+3 bits for 8 bits of data, i.e. +60 % compared with a conventional SECDED), it achieves an 80 % correction rate for adjacent multiple-bit errors (MCU), whereas traditional codes are designed to correct only a single error.

This property is particularly relevant because the reduction in technology nodes mechanically increases the probability of MCUs : the charge required to switch a transistor decreases and memory cell density increases, making it more likely that a single particle will affect several neighboring bits.

The proposed ECC therefore provides a response directly suited to this technological evolution.

Integration into the ASIC flow made it possible to quantify the PPA-F (Power, Performance, Area, Reliability) trade-off induced by each of the two protection levels :

- the addition of the ECC results in a +15 % area overhead and a +30 % power consumption overhead, as well as a frequency limitation imposed by the decoder's combinational logic chain located on the critical path at the SRAM output ;
- switching to the RHBD library (180 nm) results in significantly higher overheads ($\times 59$ in area, $\times 52$ in power consumption, frequency divided by more than 4), attributable to the intrinsic geometrical cost of radiation hardening by design (Guard Rings, ELT) and the change in technology node.

These results illustrate the complementarity of the two approaches : the ECC handles transient errors (SEU, MCU) at the logical level, while RHBD protects the silicon against cumulative and destructive effects (TID, SEL). It should also be noted that the addition of these protection mechanisms inevitably requires relaxing the conventional PPA constraints.

F.1.1 Limitations of the Current Solution and Future Developments

Several limitations should be highlighted. First, the robustness evaluation relies exclusively on simulations : no physical testing under a particle beam (protons, heavy ions) could be carried out. Furthermore, TID and SEL tolerance was only discussed qualitatively, as no circuit was fabricated.

Second, the selected ECC shows its limitations beyond two adjacent faults (correction rate reduced to 31 %), which makes it necessary to combine it with a hardened layout to remain effective under real-world conditions.

Finally, the designed cache remains deliberately small (16 lines, 16-bit words), for educational purposes ; however, the developed control and ECC logic could be directly transferred to a larger-capacity cache.

Several avenues for future development emerge from this work. At the architectural level, pipelining the ECC decoder could be considered to reduce the critical path length and improve the maximum frequency. However, it should be noted that this change would introduce an additional latency cycle for memory accesses. Hence, it would be interesting to study in greater detail the speedup provided by the pipeline.

Regarding validation, a quantitative characterization of TID and SEL (through TCAD simulation or irradiation testing) would make it possible to complete the PPA-F summary. Finally, a more in-depth comparison with other architectural hardening approaches (temporal redundancy, partial TMR) would make it possible to position the proposed ECC more precisely within the range of solutions available for space applications.

F.2 Personal Conclusion

This internship was my first professional experience truly aligned with the field in which I wish to develop my career. Furthermore, it took place as part of my international mobility in Brazil, making it a particularly meaningful experience for me.

The complete autonomy I was given on this project introduced me to a working method based on continuous research and experimentation. This way of working was somewhat unsettling at first, before I quickly came to appreciate the freedom and independence it provided.

Immersion in a structured professional environment (computing server, licensed tools, remote access) was also highly instructive. In addition, the many interactions with other laboratory members working on a wide variety of topics allowed me to discover other areas of electronics beyond the subject of my internship.

From a technical perspective, this internship allowed me to gain a concrete understanding of the physical limitations of components in a space environment, as well as the techniques developed to address them. In particular, I was able to explore hardening techniques at the layout level (ELT, Guard Rings) as well as at the architectural level (ECC).

I also learned SystemVerilog, whereas I had previously worked with VHDL. I particularly appreciated the readability provided by structures while remaining a hardware description language that is very close to the underlying hardware, without an excessive abstraction layer. The cache design also provided an opportunity to apply concretely the concepts covered in the processor architecture course.

Finally, this internship allowed me to significantly deepen my understanding of the ASIC flow as a whole. I had already had an initial exposure to it at ENSEIRB, but integrating the RHBD library into the existing Makefile enabled me to understand its operation in much greater depth.

This development of technical skills, combined with the personal experience of my stay in Brazil, is the most significant outcome of this internship for me.

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H Appendices

H.1 Cache Data Structures (SystemVerilog Package)

Listing 1 presents the three structures centralized in the `cache_pkg` package, used to decompose the CPU address and represent a cache line with and without the ECC redundancy bytes.

Listing 1: Data structures defined in the cache package

```
1 typedef struct packed {
2     logic [3:0] tag;
3     logic [2:0] index;
4     logic      offset;
5 } addr_cpu_t;
6
7 typedef struct packed {
8     logic [3:0] tag;
9     logic      valid;
10    logic [7:0] data1;    // most significant byte
11    logic [7:0] data2;    // least significant byte
12 } data_cache_t;
13
14 typedef struct packed {
15     logic [3:0] tag;
16     logic      valid;
17     logic [7:0] data1;
18     logic [7:0] redun1;  // ECC redundancy of data1
19     logic [7:0] data2;
20     logic [7:0] redun2;  // ECC redundancy of data2
21 } data_cache_ecc_t;
```

H.2 Example of ECC Encoding/Decoding on an 8-bit Word

Figure 10 gives the XOR equations used for encoding and decoding an 8-bit data word, corresponding to the transformation A and its pseudo-inverse A^{-1} presented in Section C.2.2.

$r_7 = d_5 \oplus d_1 \oplus d_3,$	$d_7 = r_7 \oplus r_3 \oplus r_2 \oplus r_5,$
$r_6 = d_0 \oplus d_7,$	$d_6 = r_7 \oplus r_6 \oplus r_5 \oplus r_3 \oplus r_2 \oplus r_1 \oplus r_0,$
$r_5 = d_5 \oplus d_3 \oplus d_2,$	$d_5 = r_7 \oplus r_5 \oplus r_4 \oplus r_1 \oplus r_0,$
$r_4 = d_0 \oplus d_6 \oplus d_2 \oplus d_5 \oplus d_1,$	$d_4 = r_6 \oplus r_4 \oplus r_3 \oplus r_0,$
$r_3 = d_5 \oplus d_3 \oplus d_0 \oplus d_7,$	$d_3 = r_7 \oplus r_6 \oplus r_5 \oplus r_4 \oplus r_3 \oplus r_1 \oplus r_0,$
$r_2 = d_0 \oplus d_1 \oplus d_5 \oplus d_3 \oplus d_2,$	$d_2 = r_6 \oplus r_5 \oplus r_3,$
$r_1 = d_1 \oplus d_4 \oplus d_3 \oplus d_2,$	$d_1 = r_7 \oplus r_6 \oplus r_3,$
$r_0 = d_0 \oplus d_6 \oplus d_3 \oplus d_2 \oplus d_1 \oplus d_4.$	$d_0 = r_7 \oplus r_6 \oplus r_3 \oplus r_2 \oplus r_5.$

(a) Encoding

(b) Decoding

Figure 10: Encoding and decoding equations for an 8-bit word

For $d = (10101100)$, these equations give $r = A(d) = (01101100)$, resulting in a 16-bit encoded word $C = (d, r) = (1010110001101100)$, stored as is in memory.

H.3 Calculation of the Voltage Drop in the RHBD Circuit (*IR Drop*)

Total current consumed by the circuit : The total power consumed by the circuit is 46,62 mW at a supply voltage of 1,62 V. Therefore, the current consumed is

$$I_{\text{total}} = \frac{46,62 \text{ mW}}{1,62 \text{ V}} \approx 29 \text{ mA}$$

Resistance of the power distribution network (M1) : The block geometry is estimated at $730 \mu\text{m} \times 730 \mu\text{m}$. Assuming a uniform current distribution over a comb of 75 horizontal Metal 1 power rails (width $w = 0,425 \mu\text{m}$), each V_{DD} power rail carries an average current of :

$$I_{\text{rail}} = \frac{29 \text{ mA}}{75} \approx 0,4 \text{ mA}$$

Each rail has a dimension ratio of $L/w = 730/0,425 = 1720$ squares. With a typical sheet resistance for Metal 1 of $R_{\square} \approx 0,095 \Omega/\square$ in this technology, the electrical resistance of a complete rail is :

$$R_{\text{rail}} = R_{\square} \times \frac{L}{w} = 0,095 \times 1720 = 163 \Omega$$

Calculation of the voltage drop (*IR Drop*) : Assuming a uniformly distributed current density along the power rail, the maximum effective voltage drop at the end of the line is expressed as :

$$V_{\text{drop}} \approx \frac{1}{2} \times I_{\text{rail}} \times R_{\text{rail}} = \frac{1}{2} \times 0,4 \text{ mA} \times 163 \Omega \approx 32 \text{ mV}$$

Conclusion on the voltage drop : It can therefore be concluded that the supply voltage drop is 2%, showing that the vertical power rails can be removed without significantly affecting the circuit's power supply (industrial reference value: $< 5\%$).

H.4 Detailed Methodology of the Fault Injection Campaigns

Python Campaign (Monte Carlo)

The software model reproduces bit by bit the encoding logic, decoding logic, and decision rule of the ECC implemented in `ECC_decoder`[2]. Each iteration follows the following protocol : random generation of an 8-bit word d , encoding into a 16-bit codeword $C = (d, r)$, application of a corruption mask, decoding/correction, and then classification of the result (success, false correction, detected but uncorrected error).

Two corruption scenarios are tested :

- **random faults** (modeling independent SEUs) : positions are independently selected among the 16 bits of the codeword ;
- **adjacent faults** (modeling MCU/MBU) : a random starting position, followed by a *burst* of N consecutive inverted bits, representative of the localized charge deposition caused by a high-energy solar particle (up to 100 MeV).

For each configuration (1 to 4 corrupted bits), 100 000 trials are performed. Since the state space of the codeword is $2^{16} = 65\,536$ combinations, this number of trials exceeds the system cardinality and ensures statistical convergence of the measured rates.

H.5 Software/Hardware Cross-Validation Methodology

The `ECC_full_decoder_tb` testbench directly instantiates the `ECC_full_decoder` module and applies a series of test vectors read from a `vector.txt` file, previously generated by the Python model (§ 4.2, Appendix H.4). Each line of the file contains five fields : the corrupted data, the corrupted redundancy, the expected value after correction, the number of injected faults, and an indicator of the fault type (random or burst).

For each vector, the testbench provides the data and redundancy as inputs to the decoder, compares the corrected output with the expected value, and then increments a success counter classified by configuration (number of faults $\times$ distribution type). Table 6 presents the complete results obtained over 10 000 vectors per configuration.

Table 6: Correction rates obtained through hardware (RTL) cross-validation

Configuration	Success / Total	Rate
1 error, random	10000 / 10000	100,00 %
2 errors, random	4308 / 10000	43,08 %
3 errors, random	1588 / 10000	15,88 %
4 errors, random	388 / 10000	3,88 %
1 error, burst	10000 / 10000	100,00 %
2 errors, burst	7970 / 10000	79,70 %
3 errors, burst	7124 / 10000	71,24 %
4 errors, burst	3061 / 10000	30,61 %

These results are directly comparable to those of the Python campaign (§ 4.2) : the observed differences (less than 1,5% for each configuration) are consistent with the differences in random sampling between the two campaigns and confirm that the RTL module faithfully reproduces the decision logic of the algorithmic model.